

A quadratic barrier for corrections in one column per row

Asymptotic research note; independently reviewed

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The polynomial exceptional-die construction starts with distinct scalar nodes and corrects at most one comparison column at each node. The following argument shows that this particular strategy cannot reach a sub-quadratic number of rows. It is not a general lower bound for dice.

Theorem 1. *Let $m \geq 3$ and let $t_1, \dots, t_K \in [0, 1]$ be pairwise distinct. Suppose a $K \times m$ array has the form*

$$p_j(q) = t_q + \delta_j(q),$$

where at every row at most one $\delta_j(q)$ is nonzero, and suppose

$$\frac{1}{K} \sum_q \prod_{j \in S} p_j(q) = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Then $K \geq cm^2$ for an absolute positive constant c . More specifically, either the scalar nodes themselves give an equal-weight interval quadrature of degree $m - 1$, or at least $m^2/2$ rows have a nonzero correction.

Proof. Put

$$b_{j,r} = \frac{1}{K} \sum_q \delta_j(q) t_q^r.$$

At a row, all terms involving two corrections vanish. For a set S of size s the mixed-moment equation is therefore

$$\frac{1}{K} \sum_q t_q^s + \sum_{j \in S} b_{j,s-1} = \frac{1}{s+1}.$$

For $1 \leq s \leq m - 1$, comparing two sets differing in one index shows that $b_{j,s-1}$ is independent of j . Denote their common value by b_{s-1} .

If all $b_0, \dots, b_{m-2}$ vanish, the scalar nodes integrate every monomial through degree $m-1$. The classical lower bound for equal-weight uniform-interval quadrature gives $K \geq cm^2$.

Otherwise let $Q_j = \{q : \delta_j(q) \neq 0\}$ and $k_j = |Q_j|$. These sets are pairwise disjoint. For any two indices $i \neq j$ the signed measure

$$\sum_{q \in Q_i} \delta_i(q) \delta_{t_q} - \sum_{q \in Q_j} \delta_j(q) \delta_{t_q}$$

annihilates all polynomials of degree at most $m-2$. Its support consists of $k_i + k_j$ distinct points and all its coefficients are nonzero. If $k_i + k_j \leq m-1$, the Vandermonde matrix would force every coefficient to vanish, which would also force all the common b_r to vanish. Consequently $k_i + k_j \geq m$ for every pair. Summing over pairs gives

$$(m-1) \sum_j k_j \geq m \binom{m}{2}, \quad K \geq \sum_j k_j \geq \frac{m^2}{2}.$$

□

The common correction functional is forced only through degree $m-2$; the full m -column equation supplies one additional aggregate condition. The proof deliberately does not assume equality of the degree- $m-1$ correction moments. Pairwise distinctness of the baseline nodes is an explicit hypothesis. The result applies to the separated-node construction in `size_bounds/polynomial_exceptional_die.tex`, but not to an arbitrary monotone comparison tensor.

For the classical scalar quadrature bound, see the uniform-weight case in Gilboa–Peled, *Chebyshev-type quadratures for doubling weights*, <https://arxiv.org/abs/1507.01505>.

Corollary 1 (Bounded overlap). *Let $t_1, \dots, t_K \in [0, 1]$ be distinct, and suppose $p_j(q) = t_q + \delta_j(q)$, $1 \leq j \leq m$, has uniform-interval mixed moments*

$$K^{-1} \sum_q \prod_{j \in S} p_j(q) = \frac{1}{|S| + 1}.$$

Suppose each row has nonzero corrections in at most d columns. There is an absolute $c > 0$ such that

$$K \geq c \min \left\{ m^2, \frac{m^{4/3}}{d^{4/3}} \right\}.$$

In particular any fixed bound on the number of corrected columns per row requires $K = \Omega(m^{4/3})$. For $d = 1$ the stronger bound $K = \Omega(m^2)$ holds.

Proof. Join two columns by an edge if some row corrects both of them. This conflict graph has at most $K \binom{d}{2}$ edges. It has an independent set of size at least

$$R \geq \frac{m^2}{m + Kd(d-1)}. \quad (1)$$

For completeness, order the vertices uniformly at random and retain those appearing before every neighbor. The retained set is independent, its expected size is $\sum_j (1 + \deg j)^{-1}$, and Cauchy-Schwarz gives (1).

On the retained columns each row corrects at most one column, and all mixed-moment identities persist. The one-column theorem therefore gives $K \geq c_0 R^2$ for an absolute $c_0 > 0$. This estimate can be made valid also for $R = 1, 2$ by reducing c_0 , since $K \geq 1$. If $Kd(d-1) \leq m$, (1) gives $K \geq c_0 m^2/4$. Otherwise (1) and the same estimate give

$$K^3[d(d-1)]^2 \geq c_0 m^4/4.$$

The claimed bound follows. When $d = 1$ use the one-column theorem directly. $\square$